

# Requirements Table

Student Startup Crowdfunding Platform — PS#40 | 5 Functional + 2 Non-Functional Requirements

| ID      | Type                       | Description                                                                                                                                                                          | Priority | Acceptance Criteria                                                                                                                                                                                                                                              | Rationale                                                                                                                                                  |
|---------|----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| FR-001  | Functional                 | Collect backer pledges and disburse funds to the student venture in tranches only upon verification of predetermined project milestones.                                             | High     | <b>Pass:</b> Milestone evidence reviewed and tranche released within 3 business days.<br><b>Fail:</b> Campaign funds disbursed without milestone sign-off.                                                                                                       | Protects backers from funding ventures that stall or misuse funds; gives faculty an accountability checkpoint.                                             |
| FR-002  | Functional                 | Allow a Campaign Backer to browse active campaigns, select a reward tier, and submit a pledge amount meeting the tier minimum.                                                       | High     | <b>Pass:</b> Pledge recorded against selected tier with a unique pledge ID; amount $\geq$ tier minimum.<br><b>Fail:</b> Pledge below tier minimum accepted, or pledge recorded with no associated tier.                                                          | Reward tiers are the core backer incentive and must be enforced consistently to keep founder fulfillment obligations well-defined.                         |
| FR-003  | Functional                 | Provide a Faculty Committee interface to review milestone evidence and record an Approve / Reject / Request-More-Info decision with comments.                                        | High     | <b>Pass:</b> Decision is timestamped, attributed to a named reviewer, and immutable; Approve triggers FR-001 tranche release.<br><b>Fail:</b> Milestone status changes without a decision record, or a decision is editable post-submission.                     | Creates an auditable trail proving fund releases are backed by verified human judgment, not automated or self-reported claims.                             |
| FR-004  | Functional                 | Allow a Student Founder to create a campaign with a funding goal, deadline, milestone plan, and reward-tier catalog, subject to eligibility validation before publishing.            | High     | <b>Pass:</b> Campaign visible only after eligibility validation succeeds and milestone totals sum to the funding goal.<br><b>Fail:</b> Campaign published without validation, or milestone totals do not reconcile with the funding goal.                        | Prevents unauthorized or ineligible campaigns from reaching backers and keeps the milestone plan mathematically consistent with pledges.                   |
| FR-005  | Functional                 | Automatically cancel a campaign and refund all collected pledges to backers if the funding goal is not met by the deadline.                                                          | Medium   | <b>Pass:</b> If pledged < goal at deadline, all pledges refunded within 5 business days with a transaction reference.<br><b>Fail:</b> Funds retained beyond 5 business days, or a partial/no refund issued without backer-approved exception.                    | Mirrors standard all-or-nothing crowdfunding norms, protects backer trust, and avoids underfunded, half-started ventures.                                  |
| NFR-001 | Performance & Security     | The payment gateway must comply with PCI-DSS, using tokenized payment authorizations so raw card/bank data is never stored or transmitted through platform servers.                  | High     | <b>Pass:</b> 95th-percentile pledge latency < 2s under simulated peak load; PCI-DSS/tokenization audit passes with no critical findings.<br><b>Fail:</b> Raw card data found in logs/DB, or peak-load latency exceeds target.                                    | Financial data breaches carry severe legal, reputational, and trust consequences; deadline-driven traffic spikes are a predictable, testable load pattern. |
| NFR-002 | Reliability & Availability | The platform must maintain $\geq 99.9\%$ uptime during any campaign's active pledge window, with durable persistence of pledge and approval records surviving a single node failure. | High     | <b>Pass:</b> Rolling 30-day uptime $\geq 99.9\%$ on pledge/approval endpoints; simulated node failure results in zero lost/duplicated records.<br><b>Fail:</b> Downtime budget exceeded in a pledge window, or a node-failure drill loses or duplicates records. | Backers pledge in bursts near deadlines; an outage then directly costs student ventures real funding, and financial records must never be silently lost.   |